

Pre-Lab Questions

High-pass Filter:

1. Given the capacitor values provided below, select two capacitors C_1 and C_2 that will be used for the high-pass filter, and calculate $n = \frac{C_2}{C_1}$ (per the instructor's writeup, different notation is used in the Sallen-Key document). Make sure to select C_1 and C_2 such that $n \geq 4$.
[22nF, 33nF, 47nF, 68nF, 100nF, 150nF, 220nF, 330nF, 470nF, 680nF]
2. For the specifications given in lab exercise 11, determine the resistor values needed for the high-pass circuit through theoretical calculation (you should have already reviewed how to obtain the quality factor calculation from the previous lab exercise, and the cutoff frequency is taken from the specifications).

$$n \geq 4 \quad \text{where} \quad n = \frac{C_2}{C_1}$$

$$C_1 = 33 \text{ nF} \quad C_2 = 220 \text{ nF}$$

$$n = \frac{220 \text{ nF}}{33 \text{ nF}} = 6.666667$$

$$m = \frac{(Q(1+n))^2}{n} = \frac{(0.956520215(1+6.666667))^2}{6.666667} = 8.0666412$$

$$200 = \frac{1}{2\pi RC\sqrt{mn}}$$

$$C_1 = C = 33\text{ nF}$$

$$C_2 = 220\text{ nF}$$

$$R_2 = mR$$

$$R_1 = R = 3288.86784\ \Omega$$

$$200 = \frac{1}{2\pi R(33\text{ nF})\sqrt{(8.0666412)(6.66667)}}$$

$$\doteq 3.3\text{ k}\Omega$$

$$200 = \frac{1}{2\pi R(2.4199624 \cdot 10^{-7})}$$

$$R_2 = mR$$

$$= (8.0666412)(3288.86784)$$

$$= 26530.11682\ \Omega$$

$$\doteq 26.5\text{ k}\Omega$$

$$\doteq 27\text{ k}\Omega \quad (15\text{ HZ})$$

$$200 = \frac{1}{R(1.520528 \cdot 10^{-6})}$$

$$R = \frac{1}{200(1.520528 \cdot 10^{-6})}$$

$$= 3288.86784\ \Omega$$

Real

$$C_1 = 33\text{ nF}$$

$$\rightarrow 33\text{ nF}$$

$$C_2 = 220\text{ nF}$$

$$\rightarrow 250\text{ nF}$$

Real

$$R_1 = 3.3\text{ k}\Omega$$

$$\rightarrow 3.272\text{ k}\Omega$$

$$R_2 = 27\text{ k}\Omega$$

$$\rightarrow 14.78\text{ k} + 11.83\text{ k}$$

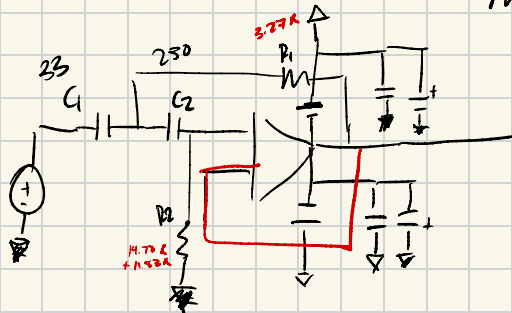
$$15\text{ k} \quad 12\text{ k}$$

$$f_n =$$

$$\frac{1}{2\pi \sqrt{C(26.61 \cdot 10^3)(3.27 \cdot 10^3)(33\text{ nF})(250\text{ nF})}}$$

$$= 187.8437967$$

$$\approx 200$$



Low-pass Filter:

3. Given the capacitor values provided below, select two capacitors C_1 and C_2 that will be used for the low-pass filter, and calculate $n = \frac{C_1}{C_2}$ (per the instructor's writeup, different notation is used in the Sallen-Key document). Make sure to select C_1 and C_2 such that $n \geq 4$.
4. [22nF, 33nF, 47nF, 68nF, 100nF, 150nF, 220nF, 330nF, 470nF, 680nF]
5. For the specifications given in lab exercise 11, determine the resistor values needed for the low-pass circuit through theoretical calculation (you should have already reviewed how to obtain the quality factor calculation from the previous lab exercise, and the cutoff frequency is taken from the specifications).

$$n = \frac{C_1}{C_2}$$

$$C_1 = 330 \text{ nF}$$

$$C_2 = 47 \text{ nF}$$

$$n = \frac{330 \text{ nF}}{47 \text{ nF}} = 7.021276$$

$$n = \frac{(Q(m+1))^2}{m} \rightarrow m = 5.492023$$

$$m = 5.492023$$

$$C_1 = 330 \text{ nF}$$

$$C_2 = 47 \text{ nF} = C$$

$$M = 5.492023$$

$$n = 7.021276$$

$$R_1 = mR$$

$$\cancel{R_1 = R}$$

$$R_2 = R$$

$$\cancel{R_2 = mR_1}$$

$$R = R_2 = 30.29532197 \approx 33 \Omega$$

$$R_1 = (5.492023)(30.29532197) = 166.3926051$$

$$\approx 180 \Omega (\text{close})$$

$$\frac{200}{18K} = \frac{1}{2\pi RC \sqrt{mn}} = f_n = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}}$$

$$R = \frac{1}{2\pi \frac{200}{18K} (47 \cdot 10^{-9}) \sqrt{(5.492023)(7.021276)}}$$

$$R = 4.31478828 \approx 30.29532197 \Omega$$

$$\approx 4.7 \Omega (\text{close}) \quad 33 \Omega$$

$$C_1 = 330 \text{ nF}$$

$$C_2 = 47 \text{ nF}$$

$$R_1 = 180 \Omega$$

$$R_2 = 33 \Omega$$